

Plots without goodwill/grievance

```
#Nation p
#a=0.1 #mutual fear,
#b=0.7 #expenditure
#r=0 #goodwill/grievance

#Nation g
#c=0.2 #mutual fear
#d=0.1 #expenditure
#s=0 #goodwill/grievance

#Simulation variables
t = 50
dt=0.1
steps=t/dt

euler = function(a,b,c,d,r,s) {
  #Matrix and initial
  ArmsRace = matrix(0,nrow=3,ncol=steps+1) #3 rows are for t,p,g, n+1 columns for including both endpoints
  ArmsRace[1,1]=0
  ArmsRace[2,1]=0.5 #p initial condition
  ArmsRace[3,1]=1 #g initial condition

  #Euler Method
  for (i in 1:steps) {
    t_i=ArmsRace[1,i]
    p_i=ArmsRace[2,i]
    g_i=ArmsRace[3,i]

    dp=a*g_i-b*p_i+r
    dg=c*p_i-d*g_i+s

    ArmsRace[1,i+1]=t_i+dt
    ArmsRace[2,i+1]=p_i+dt*dp
    ArmsRace[3,i+1]=g_i+dt*dg
  }
  return(ArmsRace)
}

#bd>ac
ArmsRace = euler(a=0.1,b=0.7,c=0.2,d=0.1,r=0,s=0)
png("bdgeac.png", width=1000, height=1000) # Saves plot as a png
par(cex=3.5)
plot(ArmsRace[1,], ArmsRace[2,], type = "l",col="blue",
xlab="Time", ylab="National Arms",
main = "a=0.1, b=0.7, c=0.2, d=0.1",
ylim=range(c(ArmsRace[2,],ArmsRace[3,])), lwd=10)
lines(ArmsRace[1,],ArmsRace[3,],col="red",lwd=10)
legend("topright",legend=c("Nation P","Nation G"),col=c("blue","red"),lwd=10)
dev.off()

## pdf
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```
##bd<ac
ArmsRace = euler(0.5,0.5,0.2,0.1,0,0)
png("bdleac.png", width=1000, height=1000)
par(cex=3.5)
plot(ArmsRace[1,], ArmsRace[2,], type = "l",col="blue",
      xlab="Time", ylab="National Arms",
      main = "a=0.5, b=0.5, c=0.2, d=0.1",
      ylim=range(c(ArmsRace[2,],ArmsRace[3,])), lwd=10)
lines(ArmsRace[1,],ArmsRace[3,],col="red",lwd=10)
legend("topleft",legend=c("Nation P","Nation G"),col=c("blue","red"),lwd=10)
dev.off()
```

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## pdf
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```
##EDGE CASE: bd=ac
ArmsRace = euler(0.7,0.5,0.5,0.7,0,0)
png("bdeqac.png", width=1000, height=1000)
par(cex=3.5)
plot(ArmsRace[1,], ArmsRace[2,], type = "l",col="blue",
      xlab="Time", ylab="National Arms",
      main = "a=0.7, b=0.5, c=0.5, d=0.7",
      ylim=range(c(ArmsRace[2,],ArmsRace[3,])), lwd=10)
lines(ArmsRace[1,],ArmsRace[3,],col="red",lwd=10)
legend("right",legend=c("Nation P","Nation G"),col=c("blue","red"),lwd=10)
dev.off()
```

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Plots with goodwill/grievances

```
# Use the euler function to look at each case depending on varying r,s values.
# Maybe create an updated euler function to cap the spending values when they turn negative
```

Plots with mutual logistic fear

```
# Create a new euler function with the logistic adjustments and fit the plots.
```

```
# Simulation variables
```

```
t = 50
```

```
dt = 0.1
```

```
steps = t / dt
```

```
euler_logistic = function(a, b, c, d, r, s, Kp, Kg, p0 = 0.5, g0 = 1) {
  # Matrix: row 1 = time, row 2 = p, row 3 = g
  ArmsRace = matrix(0, nrow = 3, ncol = steps + 1)
  ArmsRace[1, 1] = 0
  ArmsRace[2, 1] = p0
  ArmsRace[3, 1] = g0
```

```

for (i in 1:steps) {
  t_i = ArmsRace[1, i]
  p_i = ArmsRace[2, i]
  g_i = ArmsRace[3, i]

  # Logistic fear adjustment
  dp = a * g_i * (1 - p_i / Kp) - b * p_i + r
  dg = c * p_i * (1 - g_i / Kg) - d * g_i + s

  ArmsRace[1, i + 1] = t_i + dt
  ArmsRace[2, i + 1] = p_i + dt * dp
  ArmsRace[3, i + 1] = g_i + dt * dg

  # Prevent negative arms levels
  ArmsRace[2, i + 1] = max(0, ArmsRace[2, i + 1])
  ArmsRace[3, i + 1] = max(0, ArmsRace[3, i + 1])
}

return(ArmsRace)
}

# Case 1
ArmsRace = euler_logistic(a = 0.1, b = 0.7, c = 0.2, d = 0.1,
                          r = 0, s = 0, Kp = 5, Kg = 5)

png("logistic_case1.png", width = 1000, height = 1000)
par(cex = 3.5)
plot(ArmsRace[1, ], ArmsRace[2, ], type = "l", col = "blue",
      xlab = "Time", ylab = "National Arms",
      main = "a=0.1, b=0.7, c=0.2, d=0.1, Kp=5, Kg=5",
      ylim = range(c(ArmsRace[2, ], ArmsRace[3, ])), lwd = 10)
lines(ArmsRace[1, ], ArmsRace[3, ], col = "red", lwd = 10)
legend("topright", legend = c("Nation P", "Nation G"),
      col = c("blue", "red"), lwd = 10)
dev.off()

## pdf
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# Case 2
ArmsRace = euler_logistic(a = 0.5, b = 0.5, c = 0.2, d = 0.1,
                          r = 0, s = 0, Kp = 5, Kg = 5)

png("logistic_case2.png", width = 1000, height = 1000)
par(cex = 3.5)
plot(ArmsRace[1, ], ArmsRace[2, ], type = "l", col = "blue",
      xlab = "Time", ylab = "National Arms",
      main = "a=0.5, b=0.5, c=0.2, d=0.1, Kp=5, Kg=5",
      ylim = range(c(ArmsRace[2, ], ArmsRace[3, ])), lwd = 10)
lines(ArmsRace[1, ], ArmsRace[3, ], col = "red", lwd = 10)
legend("topleft", legend = c("Nation P", "Nation G"),
      col = c("blue", "red"), lwd = 10)
dev.off()

## pdf

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```
## 2
# Case 3
ArmsRace = euler_logistic(a = 0.7, b = 0.5, c = 0.5, d = 0.7,
                          r = 0, s = 0, Kp = 5, Kg = 5)

png("logistic_case3.png", width = 1000, height = 1000)
par(cex = 3.5)
plot(ArmsRace[1, ], ArmsRace[2, ], type = "l", col = "blue",
     xlab = "Time", ylab = "National Arms",
     main = "a=0.7, b=0.5, c=0.5, d=0.7, Kp=5, Kg=5",
     ylim = range(c(ArmsRace[2, ], ArmsRace[3, ])), lwd = 10)
lines(ArmsRace[1, ], ArmsRace[3, ], col = "red", lwd = 10)
legend("right", legend = c("Nation P", "Nation G"),
      col = c("blue", "red"), lwd = 10)
dev.off()

## pdf
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Logistic mutual fear with trust/hostility

```
# Simulation variables
t = 50
dt = 0.1
steps = t / dt

euler_logistic = function(a, b, c, d, r, s, Kp, Kg, p0 = 0.5, g0 = 1) {
  ArmsRace = matrix(0, nrow = 3, ncol = steps + 1)
  ArmsRace[1, 1] = 0
  ArmsRace[2, 1] = p0
  ArmsRace[3, 1] = g0

  for (i in 1:steps) {
    t_i = ArmsRace[1, i]
    p_i = ArmsRace[2, i]
    g_i = ArmsRace[3, i]

    dp = a * (1 - p_i / Kp) * g_i - b * p_i + r
    dg = c * (1 - g_i / Kg) * p_i - d * g_i + s

    ArmsRace[1, i + 1] = t_i + dt
    ArmsRace[2, i + 1] = max(0, p_i + dt * dp)
    ArmsRace[3, i + 1] = max(0, g_i + dt * dg)
  }

  return(ArmsRace)
}

# -----
# Case 1:  $bd < ac$ ,  $r < 0$ ,  $s < 0$ 
# keep other parameters same as before
# -----
ArmsRace = euler_logistic(
```

```

a = 0.7, b = 0.5, c = 0.5, d = 0.7,
r = -0.1, s = -0.1,
Kp = 5, Kg = 5
)

png("logistic_rs_case1.png", width = 1000, height = 1000)
par(cex = 3.0)
plot(ArmsRace[1, ], ArmsRace[2, ], type = "l", col = "blue",
      xlab = "Time", ylab = "National Arms",
      main = "a=0.7, b=0.5, c=0.5, d=0.7, r=-0.1, s=-0.1, Kp=5, Kg=5",
      ylim = range(c(ArmsRace[2, ], ArmsRace[3, ])), lwd = 8, cex.main = 0.6)
lines(ArmsRace[1, ], ArmsRace[3, ], col = "red", lwd = 8)
legend("topright", legend = c("Nation P", "Nation G"),
      col = c("blue", "red"), lwd = 8)
dev.off()

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## pdf
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# -----
# Case 2:  $bd = ac$ ,  $r < 0$ ,  $s > 0$ 
# keep other parameters same as before
# -----
ArmsRace = euler_logistic(
  a = 0.5, b = 0.5, c = 0.2, d = 0.1,
  r = -0.1, s = 0.1,
  Kp = 5, Kg = 5
)

png("logistic_rs_case2.png", width = 1000, height = 1000)
par(cex = 3.0)
plot(ArmsRace[1, ], ArmsRace[2, ], type = "l", col = "blue",
      xlab = "Time", ylab = "National Arms",
      main = "a=0.5, b=0.5, c=0.2, d=0.1, r=-0.1, s=0.1, Kp=5, Kg=5",
      ylim = range(c(ArmsRace[2, ], ArmsRace[3, ])), lwd = 8, cex.main = 0.6)
lines(ArmsRace[1, ], ArmsRace[3, ], col = "red", lwd = 8, cex.main = 0.4)
legend("topleft", legend = c("Nation P", "Nation G"),
      col = c("blue", "red"), lwd = 8)
dev.off()

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## pdf
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# -----
# Case 3:  $bd > ac$ ,  $r > 0$ ,  $s > 0$ 
# keep other parameters same as before
# -----
ArmsRace = euler_logistic(
  a = 0.1, b = 0.7, c = 0.2, d = 0.1,
  r = 0.1, s = 0.1,
  Kp = 5, Kg = 5
)

png("logistic_rs_case3.png", width = 1000, height = 1000)
par(cex = 3.0)

```

```

plot(ArmsRace[1, ], ArmsRace[2, ], type = "l", col = "blue",
     xlab = "Time", ylab = "National Arms",
     main = "a=0.1, b=0.7, c=0.2, d=0.1, r=0.1, s=0.1, Kp=5, Kg=5",
     ylim = range(c(ArmsRace[2, ], ArmsRace[3, ])), lwd = 8, cex.main = 0.6)
lines(ArmsRace[1, ], ArmsRace[3, ], col = "red", lwd = 8)
legend(x = 30, y = 1.2,
       legend = c("Nation P", "Nation G"),
       col = c("blue", "red"), lwd = 10)
dev.off()

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## pdf
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